

GEO-ANOM Task 1 — Field Labeling Guide

What the model detects, the decision rules behind every YES/NO call, and the specific open questions a field visit can resolve. Generated 2026-09-21.

1. What We're Tracking Right Now

Species	Coverage	Status
Poultry	358 / 389 farms	Validated: $R^2=0.655$ vs. ground truth, 99.7% precision, 0.9% true-miss rate
Swine	1 / 1 farms	Complete
Beef	3 / 3 farms	Fixed 2026-09-21 -- 2 detections added, not yet field-verified (see §4a)
Dairy	1 / 14 farms	Open -- model mostly doesn't segment dairy barn roofs at all (see §4b)
Lagoons	5 confirmed / ~10% precision at scale	Open -- full candidate list not usable as a count

Everything below either explains **why** a detection was accepted or rejected, or flags a specific farm where a real photo settles an open question.

2. Decision Rules — Buildings

The model flags a shape as a real structure using area, length, width, and aspect ratio (length ÷ width) from a pretrained poultry-house detector. These are the confirmed calls so far, from real audited examples in this project (not hypothetical).

Structure seen on the ground	Call	Why
Duck house	YES	Same long, enclosed roof signature as a broiler house (Isaac Martin)
Swine confinement barn	YES	Same signature (Grand View Farm, swine_55_lbs)
Cattle/heifer confinement barn	YES	Modern confined heifer housing shares the shape (Bistate Feeders)
Dairy free-stall / hay barn	YES, with caution	Works on 1 of 14 dairy farms only -- most dairy barn architecture does not trigger the filter at all
Round grain/feed silo	NO (correct)	Aspect ratio ~1 (circular) fails the length/width-ratio floor automatically
Warehouse / grandstand / stable (non-farm)	NO -- real false-positive class	Confirmed on Pimlico Race Course: 4 false detections from a shared long-roof shape. Removed from the Task 1 output 2026-09-21.

3. Decision Rules — Water Bodies (Lagoons)

Water body seen on the ground	Call	Why
Regular oval/rectangular pond, brownish/teal, dark liner, next to barns	YES	Confirmed real: Dulin, Todd, Tran (area 2,000-12,600 m ² , solidity 0.92-0.95)
Round/irregular pond, forest clearing, no barns on the farm	NO	Natural pond -- can still pass a solidity check locally

Water body seen on the ground	Call	Why
Water with tan/white disturbed ground around it	NO -- quarry/borrow pit	Excavated material visible, not a farm structure
Winding, only-partly-visible water body	NO -- river/tidal creek fragment	Jabar Rahim -- corrected 2026-09-21 from "uncertain" to confirmed false positive, matches this exact pattern
Small, very regular, bright blue pond near a house	NO -- residential pool	Distinguished by color channel (real lagoons are brownish/teal, not bright blue)
Stormwater retention pond on a farm property	UNRESOLVED	Flagged as a real, distinct confusion class in an earlier project meeting -- no rule built yet. If you see one, note its distance/siting relative to barns.
Dark building rooftop (metal/asphalt)	NO -- false-positive class	Can read as "water-colored" on some imagery
Forest-edge/hedgerow shadow strip	NO -- false-positive class	Solid dark canopy shadow, high enough solidity to pass some checks

4. Field Priority — Farms Worth Checking In Person

Ordered by how much a single real observation would resolve. Coordinates are the registry permit location (lat, lon), not necessarily the exact barn/lagoon.

4a. Beef — verify tonight's fix wasn't a false positive

Both farms got a new detection today from a loosened shape filter. Neither has been seen in person.

Farm	County	Coordinates	Model's box	What to check
Dwight Brandenburg/ Brandenburg Family LP	Frederick	39.40761, -77.63178	739 m², 52.5m long, 23.3m wide	Is there a real confinement barn there at all, and does it roughly match this size?
Panora Acres Inc./ Stacy Sellers	Carroll	39.69400, -76.88790	753 m², 44.0m long, 21.4m wide	Same check -- or is this a pasture-only operation with nothing built?
Bistate Feeders, LLC (reference, already confirmed)	Caroline	38.96570, -75.76886	7 boxes, 111-153m long, 12-14m wide	For comparison only -- note these are much longer/narrower than the two above. Worth a look if nearby, to sanity-check the new detections' shape against a known-real one.

4b. Dairy — the model isn't seeing these barns at all

These 4 farms produce zero raw model output, before any filter runs. What does the actual roof look like -- open-sided, curtain-wall, ridge-vented? Different material/color than a poultry house?

Farm	County	Coordinates	Headco unt	Model output
Lester C. Jones & Sons, Inc.	Kent	39.32205, -75.80897	3,300	Zero raw candidates
Horizon Organic Dairy, LLC	Kent	39.29127, -76.00922	544	Zero raw candidates
Matthew Fry/Fair Hill Farms, Inc	Kent	39.25550, -76.05751	1,050	Zero raw candidates
Oak Bluff Dairy Farms	Frederick	39.52347, -77.29351	1,280	Zero raw candidates
Oakland View Farms LLC (reference, already works)	Caroline	39.00936, -75.86744	1,465	1 box: 712 m², 70m long, 14m wide -- what does THIS barn look like, for contrast?

4c. Lagoons — one genuinely unresolved candidate

Farm	County	Coordinates	Candidate stats	Status
Paul Aaron Hutchison Sr./ Home Farm (3 candidates)	Talbot	~38.805, -76.016	4,056 / 9,057 / 11,340 m², solidity 0.86-0.92	Never resolved either way -- a direct look would close a real gap

Jabar Rahim (Dorchester) no longer needs a field check -- resolved today as a confirmed false positive (river fragment), see §3.

4d. Registry data gaps — free fixes if you're nearby

These 9 farms have real detected structures but no recorded animal type in the registry. If you can identify any of them, that's a direct fix to real data, not just model output.

Farm	County	Coordinates	Detections
Dustin Calloway/Clay Island Farm	Wicomico	38.42979, -75.78840	19
Bawi Hlun/Chan UK	Somerset	38.25310, -75.64924	9
Hannah Jones	Caroline	38.90793, -75.77524	8

Farm	County	Coordinates	Detections
Adam Stanley/Deathly Hollows	Somerset	38.02565, -75.80555	7
Eldwin Martin/Airview Farm	Wicomico	38.36011, -75.52343	7
Robert Rosado/Northwind Farm	Wicomico	38.43855, -75.62340	4
Muhammad Usman/H & N Farm, LLC	Wicomico	38.35939, -75.32408	4
Chasin Dreams Farm	Queen Anne's	39.12587, -75.90554	2
Alan & Kristin Hudson	Worcester	38.36420, -75.25488	1

GEO-ANOM • Task 1 Field Labeling Guide • generated 2026-09-21 • Full technical detail: docs/labeling_guide.md and docs/research_log.md in the GEO-ANOM-dashboard repo.